BRNO UNIVERSITY OF TECHNOLOGY
FACULTY OF INFORMATION TECHNOLOGY

Master's Thesis Assignment

169594

Institut: Department of Information Systems (DIFS)
Student: **Milostný Tomáš, Bc.**
Programme: Information Technology and Artificial Intelligence
Specialization: Computer Networks
Title: **Distributed Password-Cracking System Using the Actor Model (.NET)**
Category: Parallel and Distributed Computing
Academic year: 2025/26

Assignment:

1. Research password-cracking principles and Hashcat capabilities, focusing on workload distribution and GPU resource management. Study the actor model in .NET (e.g., Akka.NET, Orleans) and approaches for reliable distributed computation.
2. Propose an actor-based architecture with a coordinator managing jobs, work distribution, and result aggregation, and agent nodes running Hashcat instances. Define message flow, checkpointing, error recovery, and communication protocols to ensure low latency and reliability.
3. Implement a coordinator service for job submission, progress tracking, and result collection. Develop agent actors that wrap and control local Hashcat processes, handle task execution, and report results back to the coordinator.
4. Create a robust .NET wrapper around Hashcat, capable of launching tasks, managing keyspaces, parsing outputs, and handling retries or restarts. Support multiple hash modes and enforce proper GPU utilization and isolation.
5. Set up a test environment with multiple nodes (physical or virtual) and GPUs. Automate deployment, configuration, and orchestration of Hashcat agents. Use synthetic hash datasets for evaluation.
6. Measure throughput, scalability, task distribution latency, and fault tolerance. Analyze system behavior under failures and varying workloads, and summarize performance trade-offs and limitations.

Literature:

- HASHCAT PROJECT. *Hashcat Documentation.* [online]. 2025 [cit. 2025-10-12]. Available from: https://hashcat.net/wiki/
- BROWN, Anthony. *Reactive Applications with Akka. NET*. Simon and Schuster, 2019.
- BERNSTEIN, Phil, et al. Orleans: Distributed virtual actors for programmability and scalability. *MSRTR2014* 41 (2014).

Requirements for the semestral defence:

- points 1 and 2.

Detailed formal requirements can be found at https://www.fit.vut.cz/study/theses/

Supervisor: **Pluskal Jan, Ing., Ph.D.**
Head of Department: Kolář Dušan, doc. Dr. Ing.
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